Worksheet 16

October 26, 2026 | Section 5.6 Name: _____

About 10 minutes total. Begin independently, then discuss. Show your work.

1. Read a phase shift

Let

$$y = 2\sin(2x - \pi) + 1.$$

(a) Factor the argument and state the horizontal shift.

(b) State the amplitude, period, midline, and range.

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2. Build a cosine model

A sinusoid has maximum 7, minimum 1, and period 8. It is at its maximum when $x = 0$.

(a) Find the amplitude and midline.

(b) Write one equation using cosine and no horizontal shift. Show how you find the coefficient of x.